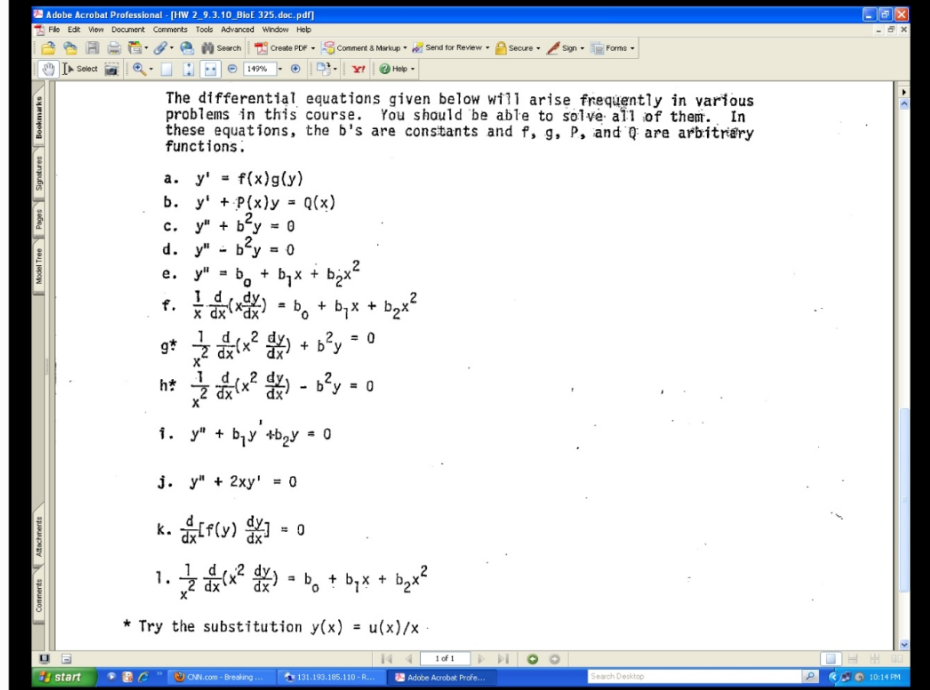


Do all problems shown below.



$$a) y' = f(x) g(y)$$

$$\frac{dy}{dx} = f(x) g(y)$$

$$\frac{dy}{g(y)} = f(x) dx$$

$$\int \frac{1}{g(y)} dy = \int f(x) dx + C$$

$$b) y' + P(x)y = Q(x)$$

$$u(x) = e^{\int P(x) dx}$$

$$u(x)y' + u(x)P(x)y = u(x)Q(x)$$

$$u(x)y' = u(x)Q(x)$$

$$u(x)y = \int u(x)Q(x) dx + C$$

$$y = \frac{\int u(x)Q(x) dx + C}{u(x)}$$

$$c) y'' + b^2 y = 0$$

$$r^2 + b^2 = 0 \quad r^2 = \pm bi$$

$$y = C_1 \cos(bx) + C_2 \sin(bx)$$

$$d) y'' - b^2 y = 0$$

$$r^2 - b^2 = 0 \quad r = \pm b$$

$$y = C_1 e^{bx} + C_2 e^{-bx}$$

$$e) y'' = b_0 + b_1 x + b_2 x^2$$

$$y' = b_0 x + \frac{1}{2} b_1 x^2 + \frac{1}{3} b_2 x^3 + C_1$$

$$y = \frac{1}{2} b_0 x^2 + \frac{1}{6} b_1 x^3 + \frac{1}{12} b_2 x^4 + C_1 x + C_2$$

$$f) \frac{1}{x} \frac{d}{dx} \left(x \frac{dy}{dx} \right) = b_0 + b_1 x + b_2 x^2$$

$$\frac{d}{dx} \left(x \frac{dy}{dx} \right) = b_0 x + b_1 x^2 + b_2 x^3$$

$$x \frac{dy}{dx} = \frac{1}{2} b_0 x^2 + \frac{1}{3} b_1 x^3 + \frac{1}{4} b_2 x^4 + C_1$$

$$\frac{dy}{dx} = \frac{1}{2} b_0 x + \frac{1}{3} b_1 x^2 + \frac{1}{4} b_2 x^3 + \frac{C_1}{x}$$

$$y = \frac{1}{4} b_0 x^2 + \frac{1}{9} b_1 x^3 + \frac{1}{16} b_2 x^4 + C_1 \ln|x| + C_2$$

$$g) \frac{1}{x^2} \frac{d}{dx} \left(x^2 \frac{dy}{dx} \right) + b^2 y = 0$$

$$\begin{aligned} \frac{d}{dx} \left(x^2 \frac{dy}{dx} \right) &= -b^2 y x^2 \\ &= -b^2 u(x) x \end{aligned}$$

$$y(x) = \frac{u(x)}{x}$$

$$h) \frac{1}{x^2} \frac{d}{dx} \left(x^2 \frac{dy}{dx} \right) - b^2 y = 0$$

$$i) \quad y'' + b_1 y' + b_0 y = 0$$

$$r^2 + b_1 r + b_0 = 0 \quad r = \frac{-b_1 \pm \sqrt{b_1^2 - 4b_0}}{2}$$

$$y = C_1 \cos(bx) + C_2 \sin(bx) \quad .$$

$$j) \quad y'' + 2xy' = 0$$

$$\text{let } u = y'$$

$$u' = y''$$

$$u' + 2xu = 0$$

$$\frac{du}{dx} = -2xu$$

$$\frac{du}{u} = -2x dx$$

$$\ln|u| = -x^2 + C$$

$$u = C_1 e^{-x^2}$$

$$y' = C_1 e^{-x^2}$$

$$y = C_1 e^{-x^2} + C_2 \quad .$$

$$k) \frac{d}{dx} \left[f(y) \frac{dy}{dx} \right] = 0$$

$$f(y) = C_1$$

$$l) \frac{1}{x^2} \frac{d}{dx} \left(x^2 \frac{dy}{dx} \right) = b_0 + b_1 x + b_2 x^2$$

$$\frac{d}{dx} \left(x^2 \frac{dy}{dx} \right) = b_0 x^2 + b_1 x^3 + b_2 x^4$$

$$x^2 \frac{dy}{dx} = \frac{1}{3} b_0 x^3 + \frac{1}{4} b_1 x^4 + \frac{1}{5} b_2 x^5 + C_1$$

$$\frac{dy}{dx} = \frac{1}{3} b_0 x + \frac{1}{4} b_1 x^2 + \frac{1}{5} b_2 x^3 + \frac{C_1}{x^2}$$

$$y = \frac{1}{6} b_0 x^2 + \frac{1}{12} b_1 x^3 + \frac{1}{20} b_2 x^4 + \frac{C_1}{x} + C_2$$